IV SEMESTER	L	T	P	C
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20AM4L01 : DATA SCIENCE LAB				

COURSE OBJECTIVES:

1. Understand the Python Programming packages Python, Numpy, Scipy, Matplotlib, Pandas, statmodels, seaborn, plotly, bokeh Language.
2. To prepare data for data analysis through understanding its distribution.
3. Exposure on data processing using NUMPY and PANDAS
4. To acquire knowledge in plotting using visualization tools.
5. To understand and implement classification and Regression Model

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

CO1: Develop relevant programming abilities.

CO2: Demonstrate knowledge of statistical data analysis techniques

CO3: Exhibit proficiency to build and assess data-based models.

CO4: Demonstrate skill in Data management & processing tasks using Python

CO5: Apply data science concepts and methods to solve problems in real-world contexts and will communicate these solutions effectively

TOOLS: Python, Numpy, Scipy, Matplotlib, Pandas, statmodels, seaborn, plotly, bokeh

LIST OF EXPERIMENTS

1. Working with Numpy arrays
2. Working with Pandas data frames
3. Basic plots using Matplotlib
4. Frequency distributions
5. Averages
6. Variability
7. Normal curves
8. Correlation and scatter plots
9. Correlation coefficient
10. Regression
11. Mini Project: any data science real-time problems using sample data sets.

References:

1. Jake VanderPlas, “Python Data Science Handbook”, O’Reilly, 2016.
2. Allen B. Downey, “Think Stats: Exploratory Data Analysis in Python”, Green Tea Press, 2014.